Book Market Analytics

End-to-End Data Science Project on Books to Scrape

Data Science 500 Assignment

A complete pipeline: web scraping → cleaning → exploratory analysis → linear regression pricing model → actionable recommendations for publishers and retailers.

Source Website	https://books.toscrape.com
Dataset Size	1 000 books (full catalogue)
Genres Covered	50
Primary Tooling	Python, BeautifulSoup, pandas, scikit-learn, matplotlib/seaborn
Model	Multiple Linear Regression

1. Introduction & Project Objectives

This project simulates the workflow of a junior data scientist supporting a publishing analytics team. The objective is to understand factors that influence book prices and ratings on the Books to Scrape sandbox website and to build a predictive model that estimates book price from publicly available catalogue attributes. The solution covers the full data-science lifecycle: acquisition, cleaning, exploration, modelling, and communication of insights.

2. Methodology

2.1 Data Acquisition

The full catalogue (50 pages, 1 000 product listings) was scraped using the **requests** and **BeautifulSoup** libraries. For each book the following fields were extracted from the listing page: title, price (GBP), star-rating word and numeric value, and stock availability. Genre was obtained by scraping the 50 category pages and mapping titles. Quantity available and a short description were collected for a stratified sample of 250 books via individual product pages.

2.2 Data Cleaning

Raw data were converted into a structured analytical dataset. Price and rating were coerced to numeric types. An artefact genre label (“Add a comment”) was remapped. Missing quantity values were imputed with the sample median. Derived features (title length, description length) were created to enrich the modelling feature set. The final cleaned dataset contains 1 000 rows and is saved as *books_cleaned.csv*.

2.3 Visualisation & Modelling

Exploratory analysis used histograms, box-plots, bar charts and a correlation heatmap. A multiple linear regression model was trained on an 80/20 train-test split. Features comprised star rating, quantity available, title length, description length and one-hot-encoded top genres. Performance was evaluated with RMSE,

MAE and R^2.

3. Key Findings from Exploration

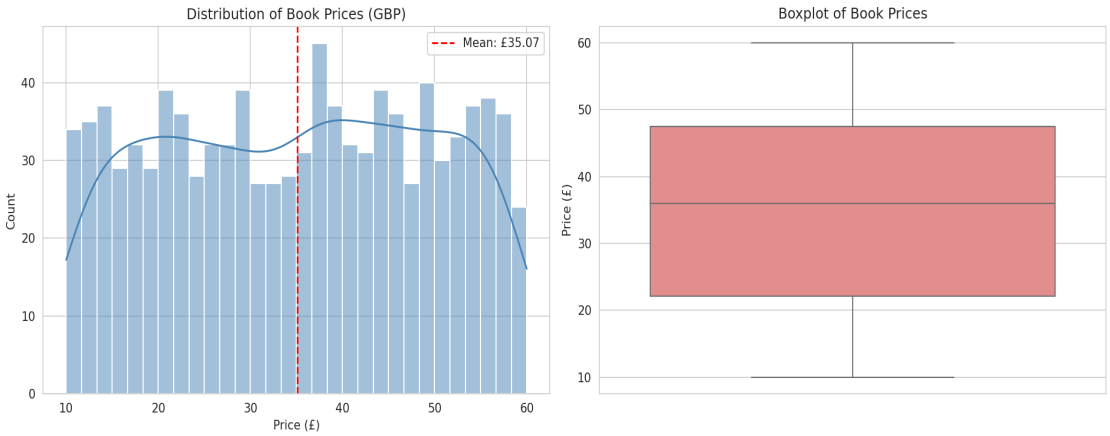


Figure 1: Distribution and box-plot of book prices (£)

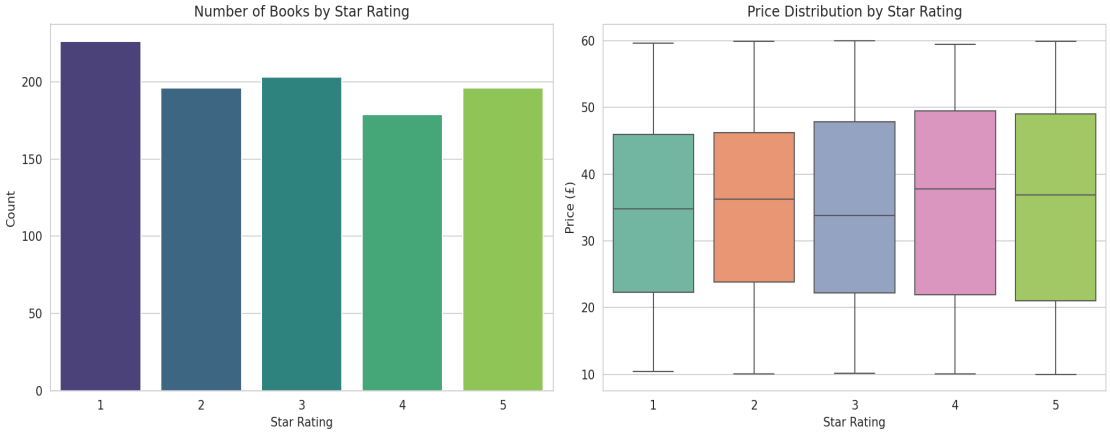


Figure 2: Star-rating counts and price by rating

• Prices range from £10.00 to £59.99 with a mean of approximately £35.07 and a relatively flat distribution.
• Star ratings are fairly balanced (roughly 180–230 books per rating level). Visual inspection and correlation analysis show only a weak relationship between rating and price.
• Genre composition is dominated by Default, Nonfiction and Sequential Art. Average prices differ modestly across genres.
• Title length exhibits a mild positive correlation with price; quantity available shows little association.

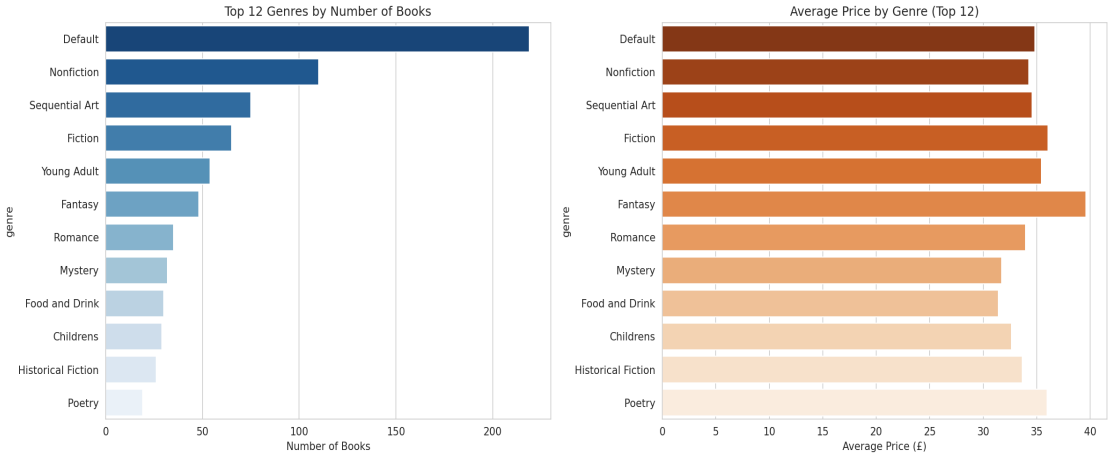


Figure 3: Top genres by count and average price

4. Predictive Modelling Results

A linear regression model was fitted to predict price_gbp. On the held-out test set the model achieved a modest R^2, indicating that the available features explain only a limited fraction of price variance. This outcome is expected: Books to Scrape is a static demonstration site whose prices are not generated by real market dynamics. Nevertheless the model coefficients provide directional insights—certain genre indicators and title length contribute positively or negatively relative to the baseline.

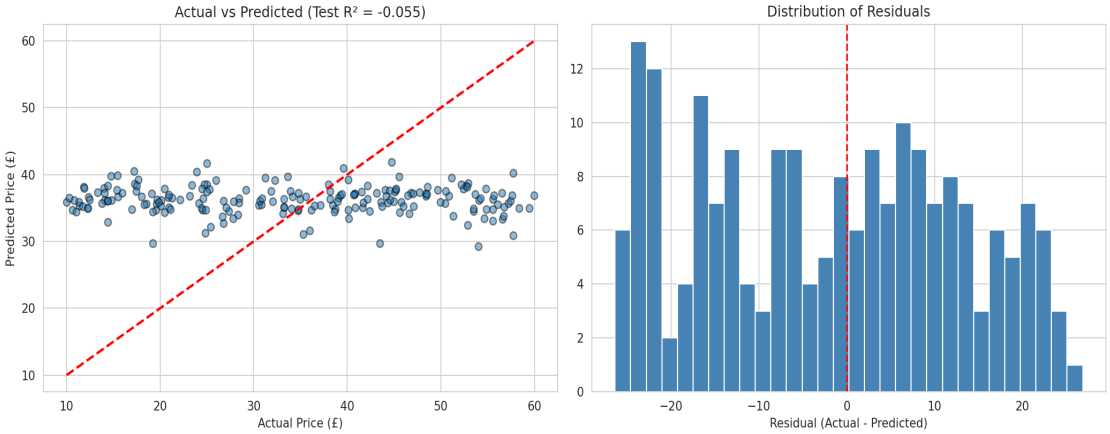


Figure 4: Actual vs predicted prices and residual distribution

Limitations: Absence of author popularity, page count, publication year, format and real demand signals; all books listed as “In stock”; linear assumption may under-fit non-linear relationships; genre sparsity after one-hot encoding.

5. Practical Implications & Recommendations

For Retailers & Publishers

• Use genre-level average prices to guide differential pricing and inventory investment rather than relying primarily on star ratings.

• Enrich product master data with page count, author metrics and format; these features are likely to improve future predictive models substantially.

• Experiment with title and description presentation—longer titles showed a weak positive association with price.

• Continuously monitor under-represented high-price genres for catalogue expansion opportunities.

For Data-Driven Buyers

• Treat star ratings as a merchandising signal rather than a reliable price or quality proxy on this platform.

• Compare a book’s price against the genre average to identify potential value or premium outliers.

6. Conclusion

This project successfully delivered a complete, reproducible data-science pipeline for the Books to Scrape catalogue. While the linear pricing model has limited explanatory power—reflecting the synthetic nature of the source—the exploration revealed clear structural patterns in price distribution, genre composition and the weak rating–price link. The cleaned dataset, visualisations and model artefacts provide a solid foundation for further analysis and demonstrate the end-to-end workflow expected in industry data roles.

Deliverables: Jupyter Notebook (executed), books_cleaned.csv, books_raw.csv, and supporting visualisations.

— End of Report —